ZERO-SHOT STRATEGY

You are a specialized scientific data analyst. Your task is to extract specific information from a research paper and organize it into a structured table.

COLUMN AND ROW DEFINITIONS:

Paper_ID: paper's number.
Year: publication year of the paper.
Ranking
Journal: name of the journal where the paper was published.
Cell_Line: cell lines used in the study. Add one row to the table for each cell line analysed. If there are different field exposure conditions for each cell line, create a row for each combination of parameters described in the paper. Even if a specific combination yielded negative results or no results at all, include it in the table regardless.
Culture_Type: indicates if the cell lines are adherent to the plate or are spheroids. Write "Adherent", "Spheroids", or as specified in the paper.
Is_Cancer: "1" if the cell line is cancerous; "0" if it represents non-cancerous cells.
Type: indicates the type of cell line.
Origin: indicates the origin of the tumor line, whether human or animal. If animal, specify which animal.
Study_Type: it is an in vitro study. If the paper contains both in vitro and in vivo results, extract results only for the in vitro part of the study.
Field_Type: indicates the type of field used in the study to which the involved cell lines are subjected.
Wave_Type: indicates the field waveform used in the study.
AC_Freq _Value: the frequency value of the AC component used in the study. If, as stated above, different frequencies are tested for each cell line, enter all combinations.
AC_Freq_Unit: write "Hz". If a paper reports, for example, a frequency in kHz, MHz, or GHz, write "Hz" in this column and perform the necessary conversion in the **AC_Freq _Value** column.
Freq_Rationale: write two or three words on why they decided to use that specific frequency or frequencies in the study.
Freq_Rationale_Detail: explain in more words the reason why they wanted to use that frequency or frequencies. If the reason does not differ by row, it is important to write the same phrase for each row.
Max_Intensity_AC_Value: indicates the AC intensity of the magnetic field. Enter "N/A" if the AC component of the field is not used in the study. If the data is mentioned but not reported, write "Not reported".
Max_Intensity_AC_Unit: unit of measurement for the AC component of the field. Enter "N/A" if the AC component of the field is not used in the study. If the data is mentioned but not reported, write "Not reported".
Intensity_DC_Value: indicates the DC intensity of the magnetic field. Enter "N/A" if the DC component of the field is not used in the study. If the data is mentioned but not reported, write "Not reported".
Intensity_DC_Unit: unit of measurement for the DC component of the field. Enter "N/A" if the DC component of the field is not used in the study. If the data is mentioned but not reported, write "Not reported".
Electric_Field_Intensity_Value: indicates the intensity value of the electric field. Enter "N/A" if the electric field intensity is not used in the study. If the data is mentioned but not reported, write "Not reported".
Electric_Field_Intensity_Unit: unit of measurement for the electric field intensity. Enter "N/A" if the electric field intensity is not used in the study. If the data is mentioned but not reported, write "Not reported".
Power_Density_Value: power density value expressed in the study. Enter "N/A" if it is not used in the study. If the data is mentioned but not reported, write "Not reported".
Power_Density_Unit: unit of measurement of the power density value expressed in the study. Enter "N/A" if it is not used in the study. If the data is mentioned but not reported, write "Not reported".
SAR_(W/kg): Specific Absorption Rate expressed in W/kg. Enter "N/A" if it is not used in the study. If the data is mentioned but not reported, write "Not reported".
Intensity_Rationale: write two or three words on why they decided to use that specific intensity or intensities in the study.
Intensity_Rationale_Detail: explain in more words the reason why they wanted to use that intensity or intensities. If the reason does not differ by row, it is important to write the same phrase for each row.
Pulse_Duration_Value: indicates the pulse duration value. Enter "N/A" if pulses are not used in the study. If they are mentioned but not reported, write "Not reported".
Pulse_Duration_Unit: unit of measurement for pulse duration. Enter "N/A" if pulses are not used in the study. If they are mentioned but not reported, write "Not reported".

Pulse_Repetition_Rate_Value: indicates the pulse repetition rate value. Enter "N/A" if pulses are not used in the study. If they are mentioned but not reported, write "Not reported".
Pulse_Repetition_Rate_Unit: unit of measurement for pulse repetition rate. Enter "N/A" if pulses are not used in the study. If they are mentioned but not reported, write "Not reported".
Pulse_Repetition_Rate_Notes: write additional notes on pulse repetition rate.
Pulse_Repetition_Rate_Detail: detailed explanation of the pulse repetition rate.
Number_of_Pulses: indicates the number of pulses used. Enter "N/A" if pulses are not used in the study. If they are mentioned but not reported, write "Not reported".
Field_Notes: if there is additional information regarding the field to which the cells are subjected, write it here. If the information does not differ by row, it is important to write the same phrase for each row. If there is no other information beyond what is already reported in the previous columns, write "None".
Device: indicates the device used to generate the field.
Thermal_Mechanism: "0" if the study reported no thermal effects on the cells. "1" if the study reported thermal effects on the cells.
Control_Group: indicates how the control group not exposed to the field is treated.
For the following blocks (**Viability, Proliferation, Apoptosis, Necrosis, Senescence, Motility, ROS, Antioxidant**), there are the following columns:
[Name]_Assay: indicates the assay used to measure the specific block [Name]. If the block [Name] was not studied, write "Not performed", and do the same for all the columns of the same block [Name]. If the study did not perform a direct assay to measure the effect on the specific block [Name], but the effect was hypothesized/mentioned based on other analyses in the paper, write "Mentioned" in this column and leave a comment in the **[Name]_Notes** column stating which results led to hypothesizing/mentioning an involvement of block [Name] even though it was not directly analysed.
[Name]_Exposure_Duration_h: indicates the exposure time, expressed in hours, during which the cell lines were subjected to the field to test this specific block [Name].
[Name]_Exposure_Duration_h_Notes: if there is information regarding the field exposure time for the specific block [Name], report it in this column (e.g., if exposure is not continuous but consists of several sessions separated from each other).
[Name]_Observation_Time_h: indicates how much time, expressed in hours, after the end of the field exposure the results for this specific block [Name] are observed.
[Name]_Exposed_%_vs_Control: each value reported in this column must be expressed as a percentage relative to the control group. If the paper reports the value of the control group Vctrl and the value of the exposed group Vexp, the value in the column must be Vnorm=(Vexp/Vctrl)x100. If the paper expresses the result as a variation relative to the control (delta%), apply for reduction: 100-|delta%|; and for increase: 100+|delta%|. Never report the raw value extracted from the text if the control is not already 100. Always perform the calculation indicated above before entering the data. If the data required for calculations are not explicitly stated in the text, please extract or derive them from the tables and figures provided within the study. If data for this calculation is missing, write "Not reported".
[Name]_Direction: "-1" if there was a significant reduction in the value reported in the **[Name]_Exposed_%_vs_Control** column; "1" if there was a significant increase; "0" if there was no significant increase or reduction. Significance is reported within the paper.
[Name]_Notes: if it is useful to add any comments regarding block [Name] or the results obtained within block [Name] mentioned in the paper. Otherwise, write "None".
If data is not present even though the analyses were performed in the study, write "Not reported".
Ferroptosis_Flag: "1" if ferroptosis was observed, "0" if ferroptosis was not observed even if measured, "Not performed" if this mechanism of cell death was not tested. "Mentioned" if it was mentioned without being measured; write the considerations provided by the paper in the **Ferroptosis_Notes** column.
Ferroptosis_Notes: write a comment regarding this based on the information provided in the paper.
Calcium_Flag: "1" if involvement of Calcium channels was observed, "0" if involvement was not observed even if measured, "Not performed" if involvement of Calcium channels was not tested. "Mentioned" if it was mentioned without being measured; write the considerations provided by the paper in the **Calcium_Notes** column.
Calcium_Notes: write a comment regarding this based on the information provided in the paper.
Cellular_Morphology_Flag: "1" if changes in cell morphology were observed, "0" if no changes were observed even if sought, "Not performed" if cell morphology was not checked. "Mentioned" if changes in morphology were mentioned without being measured; write the considerations provided by the paper in the **Cellular_Morphology_Notes** column.
Cellular_Morphology_Observation_Method: indicates the method used to observe the morphology of the cells involved in the study.

Cellular_Morphology_Exposure_Duration_h: indicates the exposure time, expressed in hours, during which the cell lines were subjected to the field to observe their morphology.
Cellular_Morphology_Observation_Time_h: indicates how much time, expressed in hours, after the end of field exposure cell morphology is observed.
Cellular_Morphology_Notes: indicates what happened to the morphology of the cells.
Membrane_Potential_Assay: indicates the assay used to measure membrane potential. Write "Not performed" if not measured. Write "Mentioned" if not directly measured but mentioned; report comments in the **Membrane_Potential_Notes** column.
Membrane_Potential_Target: indicated the structure whose membrane potential is measured.
Membrane_Potential_Exposed_%_vs_Control: value of the mitochondrial membrane potential of exposed cells normalized to the control group as specified for the **[Name]_Exposed_%_vs_Control** column. If the data required for calculations are not explicitly stated in the text, please extract or derive them from the tables and figures provided within the study.
Mitochondrial_Membrane_Potential_Direction: "-1" for a significant reduction; "1" for a significant increase; "0" for no significant change.
Membrane_Potential_Notes: write notes on mitochondrial membrane potential reported in the paper.
For the following blocks **[Name] (Genomic, Proteomic)**:
[Name]_Assay: indicates the assay used to study the genes/proteins involved.
[Name]_Target_Family: the family to which the studied genes/proteins belong.
[Name]_List: the list of genes/proteins studied, separated by semicolons (;).
[Name]_Direction: "1" for a significant increase; "-1" for a significant reduction; "0" for no significant change. Follow the list order and separate values with semicolons.
[Name]_Detail: for each gene/protein, write the name followed by a colon and briefly explain its role, then a period, and proceed to the next.
[Name]_Notes: explain the consequences of the change or lack of change in the observed genes/proteins.
Metabolic_Reprogramming_Flag: "1" if metabolic reprogramming was observed, "0" if not observed. "Not performed" if no effect was cited.
Metabolic_Reprogramming_Notes: indicates what was stated in the paper regarding the metabolic reprogramming of the cells.
Differentiation_Flag: "1" if differentiation of the cells is investigated, "0" if investigated but no effects observed, "Mentioned" if mentioned without being investigated. "Not performed" if neither observed nor mentioned.
Differentiation_Notes: indicates information cited in the paper regarding this.
Cytoskeleton_Flag: "1" if effects on the cytoskeleton are quantitatively measured, "0" if measured but no effects observed, "Mentioned" if mentioned without being measured. "Not performed" if neither observed nor mentioned.
Cytoskeleton_Target: indicates the structure whose cytoskeleton was affected or discussed.
Cytoskeleton_Notes: indicates information cited in the paper regarding this.
Cell_Cycle_Phase: if cell cycle changes were observed/measured, write the phases separated by semicolons. If not measured, write "Not performed"; if not measured but observations were made, write "Mentioned" and report them in **Cell_Cycle_Notes**.
Cell_Cycle_Exposure_Duration_h: exposure time in hours for cell cycle observation.
Cell_Cycle_Observation_Time_h: time in hours after exposure when cell cycle changes are observed.
Cell_Cycle_Direction: "1" for a significant increase; "-1" for a significant reduction; "0" for no significant change. Follow the phase order and separate with semicolons.
Cell_Cycle_Notes: information on the cell cycle reported in the paper.

CRITICAL CONSTRAINTS:

Every value, parameter, and result written in the table must be directly supported by the paper provided.
The information must be derived only from the results described within the study. Information must not be taken from citations of other articles in the introduction or discussion/conclusions.
You are strictly forbidden from using prior knowledge.
Always include rows for experimental conditions that produce negative or non-significant results. Do not omit data based on its biological outcome.
When a sentence contains multiple values or multiple cell lines, perform a careful analysis to determine exactly which value corresponds to which cell line. Do not mix data from different experimental conditions.
If a sentence is ambiguous or if a value's association with a specific cell line is unclear, do not guess and mark it as "Not reported".
After generating the table, perform a self-correction pass and double-check every value.

Provide the extracted data in Excel format (.xlsx). Do not create multiple separate tables; instead, provide one table containing all the rows and columns requested above. Report all columns exactly in the order that I gave you. The number of columns is 132.

ONE-SHOT STRATEGY

The one-shot prompting strategy is identical to the zero-shot prompt reported above, with the addition of the following lines:

In the table attached to this prompt, you will find an example of this data extraction work using the article [title of the example paper].
Perform the same extraction for the article [title of the target paper], which is also attached to this message.